V01.07 – Mirror Physics I

Mirror Physics I

Constraint Propagation and the Physical Conditions for Observerhood

From Persistent Identity to Causal Structure

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Mirror Programme – Volume I: Observerhood – Physics Support Arc

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Abstract

Mirror Theory derives observerhood from constraint, viability, bounded modelling, self-modelling and recursive reliability tracking. Earlier work in Volume I formalised constraint regimes, minimal observer thresholds and identity-continuity under governed memory update. This paper asks what physical conditions must be present for such observers to be realised in a universe like ours. It does not propose a modification of special relativity, general relativity, quantum theory or thermodynamics. Instead, it identifies physical support conditions for recursive observer-continuity: ordered causal update, finite channels of propagation, locally persistent substrates, metastable memory, thermodynamic throughput and error-corrective repair. Within this framing, the invariant speed c is not redefined or exceeded; it is treated as part of the causal constraint structure that bounds physical update in relativistic spacetime. The central result is a causal support theorem: if a local system robustly maintains observer-continuity across time, then the underlying physical regime must support compatible structures of causal ordering, local persistence, memory preservation, viability-relevant interaction and correction under perturbation. The paper thereby provides a conservative bridge from Mirror mathematics to future physical work without claiming new physics. It states compatibility requirements that any later Mirror-physical extension must satisfy before proposing measurable deviations from established theory.

Keywords: observerhood; causality; relativity; constraint; viability; memory; thermodynamics; information; identity-continuity; Mirror Theory.

Contents

1 Introduction

Mirror Theory begins from a methodological refusal: observers should not be inserted into a theory as primitive explanatory atoms where they can instead be derived from simpler structural conditions. The earlier papers in Volume I treated an observer as a local, viability-constrained system that maintains a world-model, a self-model and reliability variables that guide action under perturbation [2, 3, 4]. The mathematical arc then shifted the target from instantaneous observer predicates to persistence: an observer is not merely a configuration at one time, but a process that preserves a governed identity-state across time [5, 6, 7].

This paper asks the next question. If observerhood is a continuity-preserving process, what must physical reality provide for such a process to exist?

The answer is not that consciousness is fundamental. It is not that physics is wrong. The disciplined route is more modest. A persistent observer requires a physical regime in which certain operations are possible. States must endure for non-zero intervals. Updates must be causally ordered. Memory must be physically encoded. Perturbations must be resisted or repaired. Energy gradients must support continued local organisation. Without these conditions, the formal observer predicate may remain mathematically definable, but it is not physically realisable.

The purpose of this paper is therefore not to add a new field equation. Its purpose is to define physical support conditions for Mirror observerhood.

1.1 Central claim

The central claim is the following:

> *A persistent observer is a local recursive constraint-maintenance process. Such a process can exist only in a physical regime that supports ordered causal propagation, local substrate persistence, metastable memory, thermodynamic throughput and error-corrective update.*

This is a compatibility thesis. It states what physics must provide for a Mirror observer to be realised. It does not modify relativity, quantum theory or thermodynamics, and it does not infer observers from physics alone.

1.2 Why physics enters only now

It would be premature to jump directly from self-models to spacetime. The intervening work matters. The Volume I bridge has the following structure.

Constraint calculus: $\mathcal{C} = (S, T, I, V, \kappa)$

↓

Minimal observer thresholds: $\Sigma_L(R_L) > 0$

↓

Reliability experiments: perturbation, repair and cost

↓

Identity-continuity: governed memory incorporation

↓

Physical support conditions: causality, substrate, energy and correction

Figure 1: The bridge from Mirror mathematics to physical support conditions. The present paper concerns the final line: what kind of physical regime can realise a continuity-preserving observer.

The Physics Support Arc begins only after observerhood, minimal reliability salience and identity-continuity have already been formalised [2, 3, 4, 5, 6, 7]. This order matters because it prevents an equivocation between broad physical possibility and observerhood. A universe can contain stable matter, causal order and energy gradients without every stable subsystem being an observer. Mirror observerhood requires a particular governed organisation built on top of those physical supports.

1.3 Scope discipline

This paper is deliberately conservative. It does not argue that established physics is false. It does not propose superluminal influence, consciousness-induced collapse, observer-generated spacetime or a thermodynamic proof of mind. It asks a narrower question: given the formal observer-continuity developed in Volume I, what physical enabling conditions must exist for that continuity to be instantiated?

That framing sets the standard for later Mirror physics. A future paper may ask whether a quantity resembling constraint salience can be given a physical interpretation. This paper does not do that. It establishes the compatibility and support conditions that any such later work would need to respect.

2 Position within the Mirror Programme

The Mirror Programme is organised around a shared spine: constraint, model, reliability, repair, memory, viability, continuity and scale [1]. A paper belongs inside the programme only when that vocabulary does real work: it must identify a constraint-system, state what threatens its continuity, and explain mechanisms by which modelling, reliability, repair, memory or viability enable persistence, failure or transformation.

This paper satisfies that inclusion condition in the following way:

- **Constraint-system:** bounded physical subsystems capable of instantiating observer-like continuity.
- **Continuity threat:** causal disruption, thermal noise, memory decay, substrate instability, finite bandwidth, repair failure and uncontrolled perturbation.
- **Mechanisms:** causal ordering, localisation, metastable memory, energy throughput and error-corrective repair.
- **Contribution beyond prior Volume I papers:** the paper translates formal observer-continuity into physical support requirements.
- **Failure or overreach condition:** the paper would overreach if it claimed to replace relativity, quantum theory, thermodynamics or information theory, or if it treated support conditions as sufficient for consciousness.

The paper is therefore a support paper. It is not a foundational physics replacement. Its function is to connect the formalism of observer-continuity to the physical conditions under which that formalism can be realised.

3 Background: from constraint regimes to identity-continuity

The mathematical Mirror papers treat a local system L as embedded in a constraint regime [5]

$$\mathcal{C} = (S, T, I, V, \kappa), \tag{1}$$

where S is a state space, T is a transition structure, I is a set of invariants or quasi-invariants, V is a viability functional and κ represents bounded capacity.

A minimal observer predicate was then associated with the emergence of a world-model M_E, a self-model M_L and a reliability variable R_L whose use improves expected viability after cost [6]:

$$\Sigma_L(R_L) = \mathbb{E}[V_L \mid M_E, M_L, R_L, \pi_R] - \mathbb{E}[V_L \mid M_E, M_L, \pi_0] - K(R_L, \pi_R). \tag{2}$$

Here π_R denotes a policy that acts on reliability information, π_0 denotes a policy without such information and K denotes the cost of maintaining, evaluating and acting upon reliability.

Mirror Mathematics III extended this account to identity-continuity. The identity-state of a local system was represented as $I_L(t)$, with candidate memories governed by a commit function rather than passively accumulated [7]:

$$G(m_t, R_m, I_L(t), V_L) \in \{\text{commit}, \text{quarantine}, \text{audit}, \text{reject}\}. \tag{3}$$

Continuity across an interval $[t, t+\tau]$ was then represented by a functional

$$C_L(t, t+\tau) = \text{Cont}\left(I_L(t), I_L(t+\tau), H_{t:t+\tau}\right), \tag{4}$$

where $H_{t:t+\tau}$ is the relevant history of updates, perturbations, repairs and governed memory incorporations.

The physical question is now unavoidable. What must the world be like for $C_L(t, t+\tau)$ to be physically realisable rather than merely formally definable?

4 Physical support regimes

Definition 1 (Physical support regime)**.** *A physical support regime is a tuple*

$$\mathcal{P} = (\mathcal{M}, \preceq, g, F, D, E, N), \tag{5}$$

where $\mathcal{M}$ is a spacetime manifold, lattice, network or effective arena of localisation; $\preceq$ is an ordering relation sufficient for causal or transition ordering; g is a metric or effective causal geometry where applicable; F is the collection of relevant fields, matter states or physical degrees of freedom; D is a dynamical law or transition rule; E is the thermodynamic environment; and N is the noise or perturbation structure affecting local systems.

This definition is intentionally broad. In relativistic physics, $(\mathcal{M}, g)$ may be a Lorentzian spacetime. In a discrete computational model, $\mathcal{M}$ may be a graph or lattice with an induced update order. In a biological system, F includes molecular, cellular and neural degrees of freedom. The Mirror claim is not tied to a single substrate. It is tied to the conditions that allow persistent recursive organisation.

Definition 2 (Local physical subsystem)**.** *A local physical subsystem $L \subset \mathcal{P}$ is a bounded set of physical degrees of freedom whose internal state can be distinguished from, coupled to and perturbed by an environment over a relevant interval.*

Locality does not require isolation. An observer-like system must be open enough to sense and act, but bounded enough to preserve a distinguishable internal organisation.

Definition 3 (Physical realisability of observer-continuity)**.** *A formal observer-continuity functional $C_L(t, t+\tau)$ is physically realisable in $\mathcal{P}$ if there exists a local physical subsystem $L \subset \mathcal{P}$ and an instantiation map ϕ_t from physical states of L to the relevant Mirror structures*

$$\phi_t(X_L(t)) = (M_E(t), M_L(t), R_L(t), I_L(t), V_L(t)), \tag{6}$$

such that those structures are retained, updated and used over $[t, t+\tau]$ with robustness above a task-relevant threshold under perturbations drawn from N.

A compact robustness condition is

$$\mathbb{P}_N \left[C_L(t, t+\tau) > \theta_C \right] \geq \eta, \tag{7}$$

where θ_C is a non-trivial continuity threshold and $\eta > 0$ is a non-zero robustness threshold. The point is not that every observer must satisfy a single universal numerical value. The point is that physically realised observer-continuity must persist under some non-zero perturbation range. A pattern that exists for an infinitesimal instant without recoverability, memory or causal continuity may instantiate structure, but it does not instantiate persistent observerhood.

4.1 Minimal support conditions

A physical support regime capable of realising observer-continuity must provide at least the following conditions.

Condition	Requirement
S1. Ordered update	There must be an ordering relation sufficient to distinguish earlier, later and causally relevant update.
S2. Localisation	There must be a meaningful local subsystem whose internal state is distinguishable from, but coupled to, its environment.
S3. Finite channels	Information-bearing influence must propagate through channels sufficiently stable for prediction, memory update and correction.
S4. Metastable memory	Some physical states must persist long enough to encode memory, self-state and reliability history.
S5. Thermodynamic throughput	The subsystem must access energy gradients or free-energy resources sufficient to maintain organisation against noise.
S6. Error correction and repair	The subsystem must support mechanisms by which perturbations can be detected and corrected at costs that do not always exceed viability benefit.

Table 1: Minimal physical support conditions for the realisation of Mirror observer-continuity. These are not proposed as new laws; they are preconditions for instantiating the formal observer predicate in a physical regime.

The conditions are necessary in the support sense. They are not sufficient for observerhood. A crystal has persistence without self-modelling. A channel may transmit information without identity governance. A thermostat may regulate a variable without maintaining autobiographical continuity. Observerhood requires the structured combination formalised by the earlier Mirror papers.

5 The causal support theorem

Theorem 1 (Causal Support Theorem)**.** *Let L be a local system embedded in a physical support regime $\mathcal{P}$. Suppose L physically realises observer-continuity over an interval $[t, t+\tau]$ such that*

$$\mathbb{P}_N \left[C_L(t, t+\tau) > \theta_C \right] \geq \eta \tag{8}$$

for non-trivial thresholds θ_C and $\eta > 0$. Then $\mathcal{P}$ must support, over the relevant interval and scale, at least: (i) ordered causal update; (ii) local subsystem persistence; (iii) metastable memory; (iv) viability-relevant interaction with the environment; and (v) error-corrective dynamics whose expected benefit can exceed cost under some perturbation regimes.

Proof. Observer-continuity requires a comparison between an identity-state $I_L(t)$ and a later identity-state $I_L(t+\tau)$ through an update history $H_{t:t+\tau}$. If no ordered update exists, then $H_{t:t+\tau}$ cannot be decomposed into temporally or causally relevant transitions, so continuity cannot be non-trivially evaluated. If no local subsystem persists, there is no stable referent for L across the interval. If no metastable memory exists, self-model, memory and reliability variables cannot be physically retained long enough to guide later action. If no viability-relevant interaction exists, the viability functional V_L cannot be coupled to physical update. Finally, if no error-corrective dynamics exist, perturbations cannot be selectively resisted, and identity-continuity under non-zero noise collapses except by accidental persistence. Therefore physical realisation of $C_L(t,t+\tau) > \theta_C$ with robustness $\eta > 0$ implies the listed support conditions. □

Remark 1. *The theorem is a support theorem, not a derivation of a new physical law. It states that if persistent observerhood is physically realised, then the physical regime must support certain structures. It does not state that any physical regime possessing those structures necessarily contains observers.*

The theorem is intentionally one-directional. Its value lies in clarifying dependence. It rules out any account of observerhood that treats identity-continuity as if it could float free of causal order, substrate persistence, memory, energy and repair.

6 Causality and the invariant speed c

Special relativity replaces absolute space and time with invariant spacetime structure [15, 16]. In flat spacetime, the interval may be written

$$ds^2 = -c^2dt^2 + dx^2 + dy^2 + dz^2, \tag{9}$$

where c sets the invariant light-cone structure. In modern terms, c is not merely the speed of light as an optical phenomenon. It is the invariant speed of massless propagation and the causal speed limit of special relativity [17, 18].

Mirror Physics I does not alter this structure. It gives the structure an observer-support interpretation:

> *In a relativistic physical regime, c is part of the causal constraint architecture within which persistent observers update, remember, predict and repair.*

This is an interpretation, not a modified equation.

6.1 Why finite causal structure matters

A persistent observer must integrate past state, present input and expected future consequence. Such integration depends on an ordered relation between states. Relativistic light cones provide exactly this kind of order: not a universal Newtonian present, but a local causal structure determining which events can influence which other events.

For an event x, let $J^-(x)$ denote the causal past of x and $J^+(x)$ its causal future. Any physically realised self-model update at x can depend only on physically available information in $J^-(x)$ together with locally stored memory physically present at x. Any action taken at x can influence only $J^+(x)$.

Thus an observer is always situated inside a causal cone. It does not access a view from nowhere. It updates from bounded causal history.

Proposition 1 (Causal boundedness of self-model update)**.** *In a relativistic support regime, a physically realised self-model update $M_L(x)$ at event x can be a function only of degrees of* *freedom available within the causal past of x, together with locally stored memory physically present at x.*

Proof. If an update depended on information outside $J^-(x)$ that was not already locally encoded at x, the update would require superluminal influence relative to the spacetime causal structure. Such dependence is excluded by relativistic causality. Therefore the update is bounded by causal past and local memory. □

This proposition is ordinary relativity stated in Mirror language. Its significance is conceptual. Observerhood is not detached representation of the universe. It is a local process constrained by causal availability.

6.2 No faster-than-light implication

Nothing in the foregoing implies faster-than-light travel, signalling or computation. On the contrary, Mirror Physics I treats finite causal propagation as part of the support structure for stable observerhood. If future Mirror-physical work proposes a quantity that modifies effective propagation, it must first show compatibility with the empirical success of relativity, and it must specify a testable deviation. This paper does neither because it is not that paper.

7 Memory substrates and metastability

Observer-continuity requires memory. Memory is not abstract storage floating outside physics. It is a physical stabilisation of distinguishable states. A memory bit, neural trace, database record, magnetic domain or symbolic mark persists only because a material substrate maintains a state against noise.

This connects Mirror Theory to thermodynamics and information theory. Shannon information formalises communication under uncertainty [19, 27]. Landauer's principle connects information erasure to thermodynamic cost [20]. Bennett's analysis of reversible computation clarifies the relation between logical operations and physical dissipation [21]. Mirror Theory uses these insights without reducing observerhood to information quantity alone.

Definition 4 (Metastable memory state)**.** *A physical state m is a metastable memory state for system L over horizon τ if: (i) it is distinguishable from relevant alternatives; (ii) it is physically retained over τ with probability above a task-relevant threshold; and (iii) it can influence later policy, self-model update or reliability estimation.*

A schematic retention condition is

$$\mathbb{P}_N[m(t+\tau) \sim m(t)] \geq r_m, \tag{10}$$

where $\sim$ denotes equivalence under the memory code used by L and r_m is a task-relevant retention threshold. The expression is intentionally abstract because different substrates preserve memory differently. A neuronal trace, a written symbol and a digital register have different physics but share the relevant functional role: distinguishable persistence capable of later influence.

A mere mark is not yet self-memory. In Mirror Mathematics III, memory becomes identity-relevant only when governed into the self-model. The physical substrate supplies persistence; the governance architecture supplies identity relevance.

8 Thermodynamic throughput and viability

A persistent observer is a non-equilibrium structure. It must resist noise, maintain boundaries, repair damage and update internal state. This requires thermodynamic throughput: energy and material flows that maintain organisation locally while exporting entropy to the environment.

Let $E_L(t)$ denote available free-energy resources and $D_L(t)$ denote dissipation or maintenance cost. A minimal viability condition may be written schematically as

$$\mathbb{E}[V_L(t+\tau) \mid E_L(t), D_L(t), \pi] > \theta_V, \tag{11}$$

where π is the policy governing action, repair and memory update. This is not a new thermodynamic law. It is a viability expression. Without available resources, the observer-continuity functional cannot remain high under perturbation.

Thermodynamic support also prevents a common error. The existence of energy throughput does not make a system an observer. Flames dissipate energy. Vortices maintain patterns. Crystals preserve order. None of these facts is sufficient for Mirror observerhood. The support condition says only that a viability-constrained self-modelling system cannot persist without energetic and material means of maintaining itself.

9 Error correction and repair

The Observerhood Lab sequence showed that reliability variables are not automatically beneficial [8, 9, 10, 11, 12, 13, 14]. They become useful when they are channel-specific, actionable, cost-sensitive and positively coupled to viability [9, 10, 11, 12]. Physics supplies the substrate-level analogue. Error correction matters only when correction is possible and worth its cost.

In biological systems, repair includes metabolic maintenance, immune response, neural recalibration and behavioural avoidance. In artificial systems, repair includes diagnostics, sensor recalibration, checksum verification, memory auditing and rollback. In physical systems more generally, error correction requires distinguishable states, redundancy, detection and correction channels [22, 25].

Definition 5 (Physical repair channel)**.** *A physical repair channel for subsystem L is a process ρ that maps perturbed states toward viability-preserving states at a cost K_ρ, such that for some perturbation class N_i,*

$$\mathbb{E}[V_L \mid \rho, N_i] - \mathbb{E}[V_L \mid \varnothing, N_i] > K_\rho. \tag{12}$$

This expression is the physical analogue of the salience condition $\Sigma_L(R_L) > 0$. Reliability is not valuable because it is introspective. It is valuable when it enables repair that improves persistence after cost.

Proposition 2 (Repair relevance)**.** *A reliability variable R_L can contribute to physical observer-continuity only where it can modulate at least one action, repair, quarantine, audit or memory-governance policy whose expected viability benefit can exceed its expected cost under a relevant perturbation class.*

Proof. If R_L modulates no policy, it is epiphenomenal relative to viability. If it modulates a policy whose expected cost always exceeds expected benefit, it reduces or fails to improve viability. Only where R_L can guide a policy whose expected benefit exceeds cost can it contribute to persistence after perturbation. This is the same cost-sensitive structure used in the Mirror reliability salience condition. □

10 From physics to observerhood: a support map

The bridge can be stated as a map between Mirror mathematical objects and physical support conditions.

Mirror object	Physical support	Failure mode if absent
State space S	Physical degrees of freedom	No substrate for distinguishable states
Transition structure T	Dynamics or equations of motion	No ordered evolution
Invariants I	Conserved or quasi-conserved quantities; stable patterns	No persistence of form
Viability V	Free-energy access, boundary maintenance, task survival	No organisational preservation
Capacity κ	Finite memory, computation and bandwidth	No bounded modelling problem
World-model M_E	Sensor channels and internal representations	No predictive coupling to environment
Self-model M_L	Encoded body or system state	No self-relevant control
Reliability R_L	Diagnostics, calibration, confidence and uncertainty	No correction of corrupted self-state
Identity-state I_L	Governed memory and self-model continuity	No persistent observer across time

Table 2: Mapping between Mirror mathematical objects and physical support conditions. Physical support is necessary but not sufficient for observerhood.

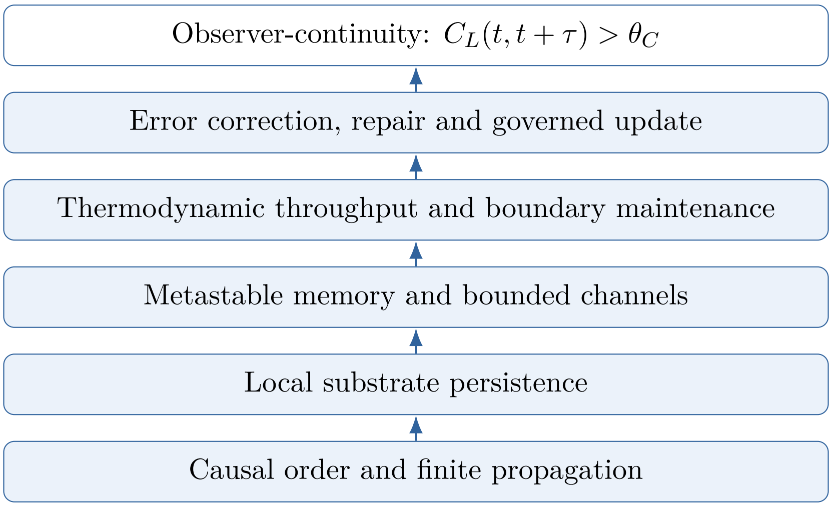

Figure 2: Support hierarchy for physical observer-continuity. The lower layers enable but do not by themselves entail the upper layer.

The map makes the core bridge explicit. Mirror Theory does not claim that physical support is sufficient for observerhood. It claims that observerhood cannot be physically realised without such support.

11 Relation to established physics

11.1 Special and general relativity

Relativity supplies causal structure [15, 16, 17, 18]. Mirror Theory treats this as a condition for local observerhood rather than an obstacle to it. In special relativity, the invariant speed c defines light cones. In general relativity, the metric field determines local causal structure in curved spacetime [17, 18].

Mirror Physics I does not modify either theory. Its claim is interpretive and structural: persistent observers are local processes whose memory, update and action are constrained by the causal geometry in which they are embedded.

11.2 Quantum theory

This paper does not derive quantum measurement, collapse, decoherence or the Born rule. It also does not claim that consciousness causes collapse. Quantum theory matters here only in the broad sense that any physical memory or computation is ultimately implemented in physical degrees of freedom. A future Mirror-physical extension would need to respect quantum constraints, including locality, no-signalling structure and decoherence where relevant [25, 26].

11.3 Thermodynamics and statistical mechanics

Thermodynamics supplies the conditions under which local organisation can be maintained away from equilibrium. Mirror observerhood requires local persistence, and local persistence in noisy environments requires energy throughput and entropy management. This aligns the framework with non-equilibrium thinking rather than with any claim that mind is thermodynamically fundamental [20, 21, 23, 24].

11.4 Information theory

Information theory supplies measures of uncertainty, channel capacity and coding [19, 27]. Mirror Theory uses these tools but does not equate observerhood with information quantity. A hard drive can store information without being an observer. A communication channel can transmit information without self-modelling. Observerhood requires reliability-governed self-models coupled to viability.

12 Compatibility conditions for future Mirror physics

A later paper may ask whether some quantity resembling constraint salience can be physically defined. This paper does not propose such a quantity as a physical field. It is useful, however, to state the discipline conditions now.

Any proposed physical Mirror quantity Φ must satisfy at least one of the following compatibility routes:

1. **Symmetry compatibility.** Φ must transform appropriately under the relevant lower-level symmetries, such as Lorentz symmetry in relativistic settings or gauge symmetry in field-theoretic settings.
2. **Coarse-grained emergence.** Φ must be shown to emerge only after coarse-graining, in a way that does not contradict lower-level symmetries.
3. **Reduction limit.** The theory must reduce to established physics in all regimes where established physics is already well tested.
4. **New prediction.** The theory must produce at least one measurable prediction that differs from standard physics.
5. **Falsifiability.** The prediction must be testable in principle, and null results must constrain the theory.

Without these conditions, a proposed Φ is not physics. It may be mathematics, metaphor or a useful modelling construct, but it is not yet a physical theory.

13 What this paper does not claim

For clarity, this paper does not claim any of the following:

- It does not claim that relativity is false.
- It does not claim that the speed of light can be exceeded.
- It does not claim that observerhood changes spacetime geometry.
- It does not claim that consciousness is fundamental.
- It does not claim that information alone creates reality.
- It does not claim that current computational labs prove physical laws.
- It does not claim that physical support conditions are sufficient for observerhood.

The claim is narrower and stronger:

> *If observerhood is a persistent recursive constraint-maintenance process, then any universe containing observers must support the physical conditions that make such processes possible.*

14 Engineering implications

Although this paper is theoretical, it suggests an engineering principle. Advanced systems should not be designed merely to store more data or use more energy. They should be designed to govern persistence under perturbation. This means engineering better commit gates, repair policies, identity governance, diagnostic reliability and resource-sensitive correction [10, 13, 14].

For autonomous probes, long-horizon artificial agents or remote robotic systems, this matters immediately. A probe sent toward another star would not merely need propulsion. It would need long-horizon self-maintenance, memory governance, error correction, identity-continuity and resource-sensitive repair under severe causal delay. In Mirror language, it would need to preserve a local observer-like control structure across time when external correction is unavailable.

This is not faster-than-light travel. It is constraint-aware engineering.

15 Toward Mirror Physics II

The next physical question is whether constraint salience can be made precise enough to interact with physical theory. A cautious future programme might ask:

1. Can a physically meaningful measure of constraint salience be defined from information, stability, causal coupling and thermodynamic cost?
2. Does such a measure transform correctly, or does it emerge only after coarse-graining?
3. Can it be related to known quantities in non-equilibrium thermodynamics, information geometry or causal network theory?
4. Does it generate any prediction not already contained in existing theory?

Only if those questions are answered should Mirror Theory propose a physical modification. Until then, Mirror Physics I remains a support-condition paper. It explains what physics must provide for observers, not how observers rewrite physics.

16 Conclusion

Mirror Theory began by deriving observerhood from viability-constrained self-modelling and recursive reliability [2, 3, 4]. The mathematics then clarified minimal observer thresholds and identity-continuity under governed memory update [6, 7]. This paper has moved the framework toward physics by asking what physical conditions make such continuity possible.

The answer is not mysterious. Persistent observers require ordered causal update, localisable substrates, metastable memory, thermodynamic throughput and error-corrective repair. In a relativistic universe, the invariant speed c belongs to the causal architecture that bounds update and interaction. Mirror Theory therefore reframes causality, memory and thermodynamic order as enabling constraints for observer-continuity.

This is not yet new physics. It is the disciplined bridge to any future physics. The next step is not to claim a new law, but to ask whether any mathematically rigorous, symmetry-compatible, empirically testable physical quantity can be associated with constraint salience. Until that is done, the strongest Mirror-physical claim is this:

> *Observers are not outside physics looking in. They are local recursive processes physically maintained by the causal and thermodynamic constraints of the universe they model.*

A Compact formal summary

A physical support regime:

$$\mathcal{P} = (\mathcal{M}, \preceq, g, F, D, E, N). \tag{13}$$

A local observer-continuity functional:

$$C_L(t, t+\tau) = \mathrm{Cont}\left(I_L(t), I_L(t+\tau), H_{t:t+\tau}\right). \tag{14}$$

A reliability salience condition:

$$\Sigma_L(R_L) = \mathbb{E}[V_L \mid M_E, M_L, R_L, \pi_R] - \mathbb{E}[V_L \mid M_E, M_L, \pi_0] - K(R_L, \pi_R) > 0. \tag{15}$$

A repair-channel condition:

$$\mathbb{E}[V_L \mid \rho, N_i] - \mathbb{E}[V_L \mid \varnothing, N_i] > K_\rho. \tag{16}$$

A robust physical realisability condition:

$$\mathbb{P}_N\left[C_L(t, t+\tau) > \theta_C\right] \geq \eta. \tag{17}$$

A causal support implication:

$$C_L(t, t+\tau) > \theta_C \Rightarrow \text{ordered update} + \text{memory} + \text{viability} + \text{correction support}. \tag{18}$$

B Glossary

Constraint A restriction on possible states, transitions or preserved structures.

Constraint regime
A structured domain of states, transitions, invariants, viability and capacity.

Observer-continuity
Preservation of a governed self-model across time under perturbation.

Causal support
: The physical ordering and propagation structure that allows local update.

Metastable memory
: A physically retained state capable of influencing later policy or self-model update.

Repair channel
: A process that maps perturbed states toward viability-preserving states at a cost.

Constraint salience
: A candidate future measure of viability-relevant structural importance; not yet a physical field.

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